

Engineering the Machine. Controlling the Process. Manufacturing the Component.

Special Purpose Machines | Machine Reconditioning & Retrofit | Precision Component Manufacturing

**SINCE****2011**

Machine building, service and manufacturing experience.

DELIVERED**300+ MACHINES**

Machines designed, manufactured and supplied to customer locations across India.

ENGINEERING**MULTI-DISCIPLINARY**

Mechanical, hydraulic, pneumatic, electrical, CNC, PLC and automation capability.

CORE MARKET POSITION

SPM Builder | Reconditioning Partner | Precision Component Manufacturer

A single engineering partner for new machine requirements, existing machine improvement and component-production needs.

Machine-tool experience applied from concept to production.

VSK Electro-Mech Solutions combines machine building, controls, field service, reconditioning and component manufacturing in one practical engineering organization.

Established in 2011, VSK Electro-Mech Solutions supports industrial customers with special purpose machines, CNC/PLC machine service and retrofit, automation, precision component manufacturing, electrical engineering and machine-tool lifecycle support. The source profile records more than 300 machines designed, manufactured and supplied, operating at customer destinations across India.

PILLAR 01

SPECIAL PURPOSE MACHINES

Customer-specific machine concepts, mechanical design, fluid-power systems, electrical controls, assembly, trials and commissioning.

PILLAR 02

RECONDITIONING & RETROFIT

Restoration and improvement of existing machines through mechanical correction, CNC/PLC upgrades, panels, drives, wiring and recommissioning.

PILLAR 03

PRECISION COMPONENT MANUFACTURING

Drawing- and process-oriented manufacturing supported by practical knowledge of CNC turning, machining centers and grinding processes.

Business Foundation

- Qualified engineering leadership with experience in machine manufacturing, repair, maintenance and quality services.
- Dedicated service engineers and skilled workers for customer execution.
- Application-oriented approach connecting machine design, controls and production requirements.
- Capability to support both standalone services and integrated machine-to-component solutions.

Engineering Foundation

- Mechanical machine design using SolidWorks.
- Hydraulic and pneumatic design, analysis and troubleshooting.
- CNC grinding-machine installation and commissioning.
- Electrical control-panel design and manufacturing to customer requirement.
- PLC/HMI and servo-controlled automation for handling, inspection and loading/unloading.



Three business pillars with equal engineering importance.

The catalog is structured around the three customer needs VSK can address independently or as a combined solution.

01

SPECIAL PURPOSE MACHINE BUILDING

Application study | Concept | Mechanical design | Hydraulics | Pneumatics | Electrical controls | CNC/PLC | Build | Commissioning

02

MACHINE RECONDITIONING & RETROFIT

Mechanical correction | Alignment | CNC/PLC upgrade | Drive and control support | Panel reconditioning | Rewiring | Recommissioning

03

PRECISION COMPONENT MANUFACTURING

Drawing review | Process planning | Turning / machining / grinding knowledge | Critical dimensions | Repeatability | Production-oriented execution

INTEGRATED CUSTOMER VALUE

One team can understand the component, the machine and the control system.

This combination helps VSK approach a requirement as a complete production problem - machine structure, spindle/interface, hydraulics, pneumatics, electrical controls, CNC/PLC logic, workholding and process execution.

When a New Machine Is Needed

- New SPM development for customer-specific production operations.
- Upgradation of existing CNC, PLC and conventional machines.
- Automation around machine loading, unloading, transfer and inspection.
- Application-specific fixtures, interfaces and machine-side support.

When Production Must Be Improved

- Precision component requirements linked to machine capability and process selection.
- Machine-tool process support where dimensional control and repeatability are critical.
- Lifecycle service after machine commissioning or retrofit.
- Electrical and mechanical industrial spares integrated with service requirements.

Mechanical, hydraulic and pneumatic design - highlighted as core capability.

VSK brings the physical machine, fluid-power systems and control architecture together as one coordinated engineering scope.

MECHANICAL

MACHINE DESIGN

Mechanical machine design using SolidWorks; fabricated and cast machine concepts; installation, alignment correction and reconditioning.

HYDRAULIC

SYSTEM DESIGN

Hydraulic design, system analysis, fault finding and problem solving for machine-tool and SPM applications.

PNEUMATIC

SYSTEM DESIGN

Pneumatic circuit/application design, system analysis, problem solving and associated air-pipeline work.

ELECTRICAL

MACHINE ELECTRIFICATION

Machine control circuits, electrical harnessing, industrial wiring, control-panel design/manufacturing and panel reconditioning.

INTEGRATION

MACHINE BUILD & COMMISSIONING

Assembly, machine installation, CNC grinding-machine commissioning, alignment and customer-site implementation.

PROCESS

APPLICATION ENGINEERING

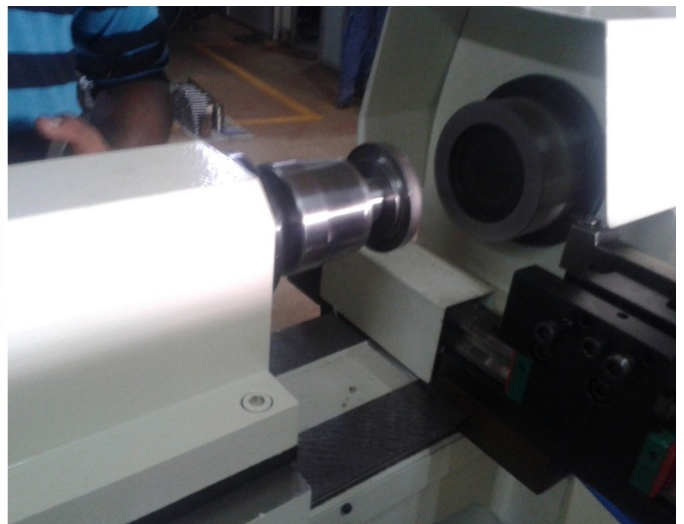
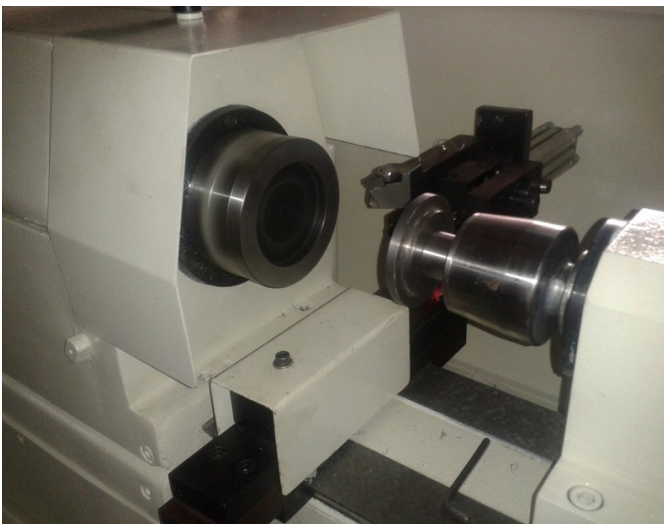
Machine concepts and support developed around the component, operation, tolerance, handling and production requirement.

Mechanical Design & Build

- Fabricated / cast machine structures and assemblies.
- Machine installation and geometric / alignment correction.
- SPM and multi-spindle machine installation and maintenance.
- Custom machine-side mechanisms and handling arrangements.

Hydraulic & Pneumatic Engineering

- Hydraulic and pneumatic systems coordinated with machine sequence and application.
- Actuation, clamping, slides and utility requirements integrated into the machine concept.
- Troubleshooting and improvement of existing fluid-power systems.
- Air-pipeline and machine utility integration.



CNC, PLC, HMI and servo control are critical engineering strengths.

Controls are treated as part of the machine architecture - not as an add-on after the mechanical build.

CNC CONTROL SYSTEMS Experience referenced with Fanuc, Siemens / Sinumerik, Heidenhain, Grundig and DMG Electro Pilot machine-control environments.	PLC CONTROL SYSTEMS PLC systems referenced include GE Fanuc, Siemens, Messung and Mitsubishi.	DRIVES MOTION / DRIVE SUPPORT CNC drive-system experience referenced with Fanuc and Siemens 610 / 611 series.
HMI OPERATOR INTERFACE PLC/HMI programming and machine operator-interface integration for production and automation sequences.	SERVO AUTOMATION Servo-controlled automation for pick-and-place, gantry loading/unloading and controlled machine-side movements.	PANELS CONTROL PANELS Machine control-panel design, manufacture, wiring, troubleshooting and reconditioning for CNC/PLC applications.

CONTROL PHILOSOPHY

CNC + PLC + Drive + HMI + Machine Sequence

VSK can support the interaction between machining control, PLC logic, machine auxiliaries, operator interface, drive systems and automation - an important advantage in SPM building and reconditioning.

Automation Applications - Pick-and-place automation. - Vision inspection integration. - Gantry loading and unloading systems. - Conveyor and machine interfacing. - Servo-controlled machine mechanisms.	Machine Control Support - Machine control-circuit fault rectification. - CNC/PLC system maintenance and troubleshooting. - Control-panel wiring and machine electrification. - CNC grinding-machine installation and commissioning. - Existing-machine sequence and controls improvement.
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Special Purpose Machines - from requirement to commissioned equipment.

Customer-specific machine building is one of VSK's primary business pillars, supported by multi-disciplinary engineering and field execution.

Application Study	Concept & Layout	Mechanical Design	Hydraulic / Pneumatic	Electrical / CNC / PLC	Manufacture & Assembly	Trials & Commission
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Machine Engineering

- Application study based on component and operation.
- Custom mechanical machine design using SolidWorks.
- Fabricated or cast machine-bed / structure concepts as required.
- Hydraulic and pneumatic systems designed around machine sequence.
- Spindle, workholding and machine-side interface integration as applicable.

Controls & Execution

- Electrical panel and machine wiring.
- CNC / PLC / HMI and drive integration.
- Servo automation, handling and inspection interfaces.
- Build, assembly, testing and trials.
- Installation, commissioning and after-sales support.

APPLICATION-FIRST DESIGN

The machine is developed around what the customer must produce.

VSK's executed references include turning, CNC boring, finishing, grinding-support, material-handling, punching / Z-cut and process-agitation equipment.



Machine reconditioning and retrofit - restoring capability and improving control.

VSK supports existing machine tools through mechanical, electrical and controls work, from fault rectification to recommissioning.

MECHANICAL RECONDITIONING Machine inspection, installation support, alignment correction, mechanical restoration and machine-side improvement.	CNC / PLC RETROFIT CNC and PLC system service, troubleshooting, upgrade / retrofit support and recommissioning.	ELECTRICAL PANEL RECONDITIONING Control-panel reconditioning, machine rewiring, control-circuit correction and electrical installation.
DRIVE MOTION SUPPORT Drive-system troubleshooting and support in Fanuc and Siemens environments referenced in the source profile.	GRINDING MACHINE-TOOL SUPPORT Service/retrofit exposure on gear grinding, ID grinding, OD grinding and centerless grinding machines.	SERVICE LIFECYCLE SUPPORT Installation, maintenance, troubleshooting, spare support and customer-site implementation.

Machine Types Supported - CNC turning machines. - VMC machines. - CNC gear grinding machines. - Gear grinding machines. - ID grinding machines. - OD grinding machines. - Centerless grinding machines.	Associated Service Capability - SPM and multi-spindle machines. - Machine control panels. - PLC/CNC system troubleshooting. - Industrial electrical installation and cabling. - HT/LT control-panel service and AMC. - Transformer / substation / hoist maintenance references.
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Precision component manufacturing with machine-tool process discipline.

The manufacturing pillar is strengthened by VSK's understanding of machine construction, CNC control, grinding processes, alignment and production engineering.

HIGH-PRECISION APPROACH

The component, process and machine are treated as one system.

Critical dimensions, machine condition, spindle/workholding interface, process sequence and repeatability are considered together so the production method is practical and controlled.

DRAWING REQUIREMENT REVIEW Customer drawing / sample, critical dimensions, material, quantity and acceptance needs form the basis of process planning.	PROCESS MACHINE-TOOL KNOWLEDGE Practical knowledge of CNC turning, machining-center and grinding-machine environments supports manufacturing decisions.	PRECISION DIMENSIONAL CONTROL Production focus on critical features, repeatability, stable processing and inspection against agreed requirements.
WORKHOLDING SETUP & INTERFACE Machine-builder knowledge helps evaluate fixture, workholding, spindle and machine-interface considerations.	IMPROVEMENT PROCESS ENGINEERING Where required, component needs can be connected to dedicated fixtures, automation or SPM development.	SUPPORT PRODUCTION LIFECYCLE Retrofit and service capability can support the equipment used to produce the component after production begins.

Manufacturing Scope - Precision components to customer drawing and technical requirements. - Turning / machining / grinding process understanding. - Application of machine-tool knowledge to manufacturability and production planning. - Customer-specific support where process and machine capability are closely linked.	Executed Precision References - 0.02 mm tolerance is recorded in the executed Z-cut machine reference. - 20-micron alignment / flatness requirements are recorded in the grinding-wheel unit reference. - +/-0.1 mm application tolerances are recorded in the table-top lathe and CNC rod-boring references. - 12-second component cycle time is recorded in the metal-facing machine reference.
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Spindle engineering and manufacturing for machine-tool applications.

VSK's spindle capability complements its SPM building, precision manufacturing and machine reconditioning activities.

NEW BUILD

CUSTOM SPINDLE UNITS

Application-specific spindle units and assemblies can be developed / manufactured to suit machine-tool and special-purpose machine requirements.

INTEGRATION

MACHINE INTERFACE

Spindle mounting, drive/interface, workholding or tooling interface, machine alignment and control integration are coordinated with the machine concept.

SERVICE

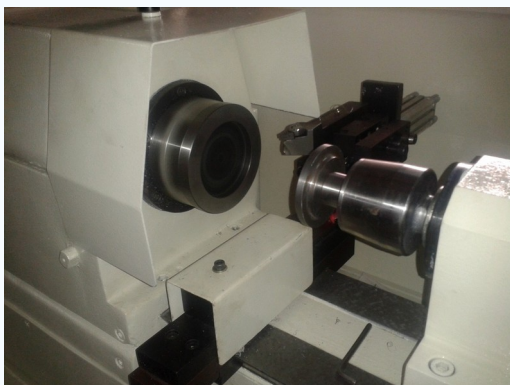
RECONDITIONING SUPPORT

Spindle-related machine service, alignment, replacement and reconditioning support can be integrated with wider machine refurbishment work.

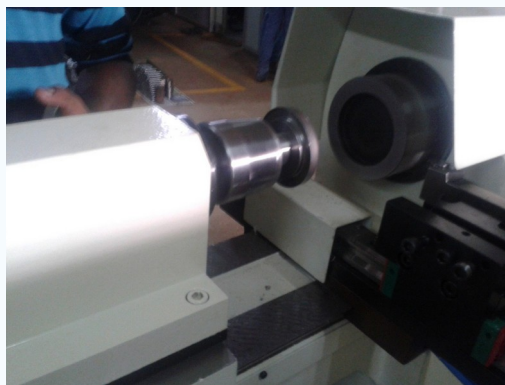
WHY SPINDLE CAPABILITY MATTERS

The spindle is central to machine accuracy, process stability and component quality.

VSK can approach spindle requirements in the context of the complete machine - structure, drive, mounting, alignment, workholding/tool interface, CNC/PLC sequence and production process.



Machine-tool spindle / workholding interface detail from the project photo set.



Representative close-up showing machine-side spindle / component interface.

Executed SPM references - turning and CNC boring.

Selected machines from the original VSK project portfolio are retained as proof of application-oriented design and build capability.

TABLE TOP LATHE

Customer: Trelleborg Sealing Systems - Hosur Road

APPLICATION

Pre-turning of PTFE rubber rods, including 850 mm length and diameters from 12 mm to 120 mm, with a target pre-turning diameter tolerance of ± 0.1 mm.

ENGINEERING / DESIGN

Fabricated bed; servo-controlled machine; PLC and touch-screen programmed operation.



ROD BORING CNC MACHINE

Customer: Trelleborg Sealing Systems - Hosur Road

APPLICATION

CNC pre-boring application for PTFE rubber tubes; the source project reference records a ± 0.1 mm tolerance requirement.

ENGINEERING / DESIGN

Cast bed; Siemens 802D 2-axis CNC controller; spindle controlled by VFD; 250 mm chuck; 5 HP spindle.

Finishing and grinding-support machines.

References demonstrating finishing-cycle design and precision support for grinding-machine applications.

METAL FACING MACHINE

Customer: Trelleborg Sealing Systems - Hosur Road

APPLICATION

Final finishing operation for metal oil seals after pressing; the source project reference records a 12-second cycle time per component.

ENGINEERING / DESIGN

Cast bed; pneumatic sliding axis; MT4 tailstock with 180 mm stroke; 5 HP spindle.



WHEEL BALANCING / ALIGNMENT UNIT

Customer: Trelleborg Sealing Systems - Hosur Road

APPLICATION

External alignment of a 1000 mm diameter grinding wheel, with a 20-micron face alignment requirement recorded in the source profile.

ENGINEERING / DESIGN

Fabricated structure; 20-micron flatness reference for wheel mounting on the wheel spindle.

Automation, handling and custom process equipment.

Machine-side automation and special equipment engineered around loading, feeding, process and tolerance requirements.

CENTERLESS GRINDING CONVEYOR

Customer: TMI Rollon Engineering Pvt. Ltd.

APPLICATION

Automatic loading and feeding of spool components to centerless grinding machines, with PLC interfacing to grinding equipment.

ENGINEERING / DESIGN

Fabricated structure; geared motor and VFD-controlled drive.



Z-CUT MACHINE

Customer: Trelleborg Sealing Systems - Hosur Road

Turcite punching / Z-cut application; source reference records 0.02 mm tolerance.



PAINT AGITATING MACHINE

Custom process equipment reference

Paint agitating machine retained from the original executed-machine portfolio.

Machine-tool, SPM and industrial engineering capability range.

A consolidated view of the machine types, custom equipment and support activities represented in the source profile and executed-machine references.

Machine-Tool Service / Retrofit Range

- CNC turning machines.
- Vertical machining centers (VMC).
- CNC gear grinding machines.
- Gear grinding machines.
- ID grinding machines.
- OD grinding machines.
- Centerless grinding machines.
- SPM and multi-spindle machines.

Executed / Custom Machine Range

- Table top lathe.
- CNC rod boring machine.
- Metal facing / finishing machine.
- Grinding wheel balancing / alignment unit.
- Centerless grinding conveyor / feeder.
- Z-cut / punching machine.
- Paint agitating machine.
- Pick-and-place, vision and gantry automation.

SPM

CUSTOM MACHINES

Design, manufacture, assembly and supply of special-purpose machines and equipment.

RETROFIT

EXISTING MACHINES

Machine reconditioning, CNC/PLC retrofit, control-panel restoration, alignment and recommissioning.

COMPONENTS

PRECISION MANUFACTURING

Precision components manufactured to customer drawing / process requirement.

SPINDLES

MACHINE-TOOL UNITS

Custom spindle units / assemblies integrated with machine-tool and SPM requirements.

AUTOMATION

PLC / HMI / SERVO

Handling, loading/unloading, conveyors, vision inspection and machine interfacing.

ELECTRICAL

INDUSTRIAL SUPPORT

Control panels, machine electrification, cabling, maintenance, HT/LT and associated electrical work.



Controls, facilities and field-service capability.

A compact engineering organization with practical machine-tool, controls and industrial electrical experience.

Controls & Automation Experience

- PLC: GE Fanuc, Siemens, Messung, Mitsubishi.
- CNC: Fanuc, Siemens / Sinumerik, Heidenhain, Grundig, DMG Electro Pilot.
- Drives: Fanuc and Siemens 610 / 611 series.
- PLC/HMI programming and servo-controlled automation.
- CNC / PLC control-panel wiring and troubleshooting.

Engineering Disciplines

- Mechanical machine design using SolidWorks.
- Hydraulic and pneumatic design / troubleshooting.
- Electrical control-panel design and manufacturing.
- Machine electrification, harnessing, cabling and terminations.
- Installation, commissioning, alignment and maintenance.

FACILITY

1,700 SQ FT

Factory covered area recorded in the source company profile.

POWER

40 HP

Connected power recorded in the source company profile.

TEAM

MULTI-DISCIPLINARY

Managers, electricians, fitters and PLC/CNC programming capability are recorded in the source profile.

FIELD SUPPORT

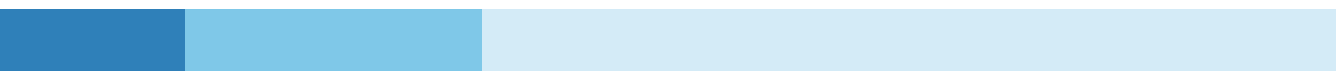
Machine installation, commissioning, maintenance and problem-solving are part of the operating model.

The original profile also records HT/LT panel work, industrial cabling, transformer / substation maintenance, lighting, hoist maintenance and associated electrical installation services.



Customer references and vendor-ready positioning.

VSK's original profile records work with established industrial customers in hydraulics, sealing systems, gears and engineered manufacturing.



CUSTOMER YUKEN INDIA LIMITED Hydraulic components reference.	CUSTOMER TRELLEBORG SEALING SOLUTIONS Metal seals, SPMs, jigs & fixtures, CNC boring machine and multiple executed-machine references.
CUSTOMER TRANSTECH GEARS INDIA PVT. LTD. Gear / engineering customer reference in the source profile.	CUSTOMER TMI ROLLON ENGINEERING PVT. LTD. Centerless grinding conveyor and material-handling reference.

Why VSK Can Be Considered - 300+ machines recorded as designed, manufactured and supplied. - Machines stated as running at customer destinations across India. - Experience across new machine build and existing-machine lifecycle support. - Integrated mechanical, fluid-power, electrical and control capability.	Typical Enquiry Areas - New special purpose machine development. - Machine reconditioning / retrofit and service. - Precision component manufacturing. - Spindle and machine-tool engineering requirements. - Automation, controls and industrial electrical support.
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VENDOR EMPANELMENT

Engineering credentials and supporting documents can be shared as required.

The profile is structured for customer introductions, vendor registration discussions, technical enquiries and website / digital business presentation.

Discuss your machine, retrofit or component requirement.

VSK Electro-Mech Solutions welcomes enquiries for special purpose machines, precision manufacturing, machine reconditioning, spindle requirements, CNC/PLC automation and engineering support.

VSK ELECTRO-MECH SOLUTIONS

SPM | RECONDITIONING | PRECISION COMPONENTS | SPINDLES | CNC / PLC | AUTOMATION

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ENGINEERING FOR PRODUCTION

Build the machine. Improve the machine. Manufacture the component.

Mechanical Design | Hydraulic Design | Pneumatic Design | Spindle Engineering | CNC / PLC | Precision Machine-Tool Processes

